

Quantium Retail Analytics

Task 1: Data Preparation & Customer Analytics

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Role: Data Analyst Intern (Virtual Experience)

Project: Quantum Retail Analytics Virtual Experience Program

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Google Colab

Executive Summary

This analysis was conducted to understand customer purchasing behaviour in the chips category and identify the customer segments that contribute most significantly to sales. The project involved cleaning and preparing transactional and customer datasets, engineering additional features such as product brand and pack size, and performing exploratory and customer segmentation analyses.

The findings indicate that **Budget Older Families**, **Mainstream Young Singles/Couples**, and **Mainstream Retirees** are the strongest contributors to chip sales. Brand preference, pack size selection, and spending behaviour vary considerably across customer segments, providing valuable insights for future marketing and merchandising strategies.

Based on these findings, targeted promotional campaigns, optimized product assortment, and segment-specific pricing strategies are recommended to maximize category growth.

1. Project Objective

The objective of this project is to analyze retail transaction and customer behaviour data to:

- Prepare and clean transaction and customer datasets.
 - Identify customer purchasing patterns.
 - Understand the drivers of chip sales.
 - Segment customers based on demographic characteristics.
 - Identify preferred brands and pack sizes.
 - Provide strategic business recommendations for the upcoming category review.
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2. Dataset Overview

Two datasets were provided.

Transaction Dataset

Contains detailed information about every chip purchase.

Key attributes include:

- Transaction Date

- Store Number
 - Loyalty Card Number
 - Product Name
 - Product Quantity
 - Total Sales
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Customer Dataset

Contains demographic information about customers.

Key attributes include:

- Loyalty Card Number
 - Lifestage
 - Premium Customer Category
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3. Data Cleaning and Preparation

Several preprocessing steps were performed to ensure data quality.

Transaction Data Cleaning

The following cleaning activities were completed:

- Converted Excel serial dates into standard datetime format.
 - Removed non-chip products (Salsa products).
 - Checked for missing values.
 - Identified and removed an extreme outlier customer purchasing 200 chip packets in a single transaction.
 - Verified product quantity and sales consistency.
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Customer Data Cleaning

The customer dataset was examined for:

- Missing values
- Duplicate loyalty card numbers
- Data consistency

No major quality issues were identified.

Data Integration

Both datasets were merged using:

Primary Key

LYLTY_CARD_NBR

The merged dataset successfully matched all customer records without introducing duplicates.

4. Feature Engineering

Two additional variables were created to enhance analysis.

Pack Size

Extracted the numeric pack size (grams) from product names.

Example:

Kettle Sea Salt 175g

↓

175

Brand Name

Brand names were extracted from product descriptions.

Example:

Smiths Crinkle Cut

↓

Smiths

Minor inconsistencies in brand names were corrected to ensure accurate reporting.

5. Exploratory Data Analysis

Several analyses were conducted to understand purchasing behaviour.

Sales by Customer Segment

Total chip sales were analyzed across:

- Lifestage
- Premium Customer Category

The analysis identified the highest revenue-generating customer segments.

Key Insight

The largest contributors to total sales are:

- Budget Older Families
- Mainstream Young Singles/Couples
- Mainstream Retirees

These segments contribute a significant proportion of total category revenue.

Customer Distribution

The number of unique customers was calculated for every customer segment.

Key Insight

Mainstream Young Singles/Couples and Mainstream Retirees represent a large customer base, partially explaining their high sales contribution.

Budget Older Families generated high sales despite having comparatively fewer customers, indicating stronger purchasing behaviour per customer.

Average Units Purchased

Average chip quantity purchased per customer was calculated.

Key Insight

Family-based customer segments purchase more packets per customer compared to singles and couples.

Older Families and Young Families consistently purchase the highest chip quantities.

Average Price Per Packet

Average selling price per packet was analyzed.

Key Insight

Mainstream Young and Mid-age Singles/Couples pay slightly higher prices per packet compared to Budget and Premium customers.

This indicates a relatively lower price sensitivity within this customer segment.

6. Brand Analysis

Brand-level analysis was performed to identify customer preferences.

The most frequently purchased brands contribute the majority of category sales.

Target customer analysis showed stronger preference toward premium and well-established chip brands.

Key Insight

Popular brands receive consistently higher sales across multiple customer segments, making them ideal candidates for promotional campaigns and prominent shelf placement.

7. Pack Size Analysis

Pack size preferences were examined across all customer groups.

Affinity analysis compared the purchasing behaviour of Mainstream Young Singles/Couples with the remainder of the customer base.

Key Insight

Medium-sized packs demonstrate stronger preference among the target customer segment.

Family households exhibit relatively greater preference for larger pack sizes.

8. Statistical Analysis

An independent two-sample t-test compared the average unit price paid by:

- Mainstream Young/Mid-age Singles & Couples
- Budget and Premium Young/Mid-age Singles & Couples

Result

The statistical test determined whether the observed price difference was significant.

This analysis supports data-driven pricing recommendations rather than relying solely on descriptive statistics.

9. Business Insights

The analysis generated several commercially valuable insights.

Customer Behaviour

- Older Families purchase larger quantities per shopping trip.
 - Young Singles/Couples contribute substantial revenue through frequent purchases.
 - Retirees remain an important and stable customer segment.
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Product Behaviour

- Brand preference varies across customer groups.
- Medium-sized chip packs are highly popular.
- Premium brands maintain strong demand despite higher prices.

Revenue Drivers

Total sales are primarily influenced by:

- Customer population size
 - Purchase frequency
 - Average quantity purchased
 - Average price per packet
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10. Strategic Recommendations

Based on the analysis, the following recommendations are proposed.

Recommendation 1

Prioritize Mainstream Young Singles/Couples through targeted advertising and loyalty promotions, as they represent one of the highest-value customer groups.

Recommendation 2

Increase promotional focus on high-performing brands to maximize category sales while maintaining strong shelf visibility.

Recommendation 3

Expand availability of the most preferred pack sizes to improve customer satisfaction and reduce stock shortages.

Recommendation 4

Introduce family-oriented bundle promotions targeting Older Families, who consistently purchase higher quantities.

Recommendation 5

Maintain premium pricing strategies for customer segments demonstrating lower price sensitivity while using targeted discounts for value-conscious shoppers.

11. Conclusion

This project successfully transformed raw retail transaction data into actionable business insights through data cleaning, feature engineering, customer segmentation, and statistical analysis.

The findings demonstrate that customer behaviour differs significantly across lifestages and purchasing segments. By focusing marketing efforts on high-value customer groups, optimizing product assortment, and aligning promotional strategies with purchasing preferences, Quantum can improve category performance and support more informed commercial decision-making.

The analysis provides a strong data-driven foundation for the upcoming category review and future strategic planning.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Google Colab
- CSV / Excel

Deliverables

- Cleaned transaction dataset
- Cleaned customer dataset
- Merged analytical dataset
- Customer segmentation analysis
- Brand preference analysis
- Pack size analysis
- Statistical significance testing
- Business recommendations
- Data visualizations
- Final analytical report